Lin BAI

Résumé

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Education

Aug. 2017 – Now **Ph.D Student**, *Worcester Polytechnic Institute*, Worcester, United States.
Major: Electrical and Computer Engineering
Research Interest: Autonomous Driving, Deep Learning Hardware Acceleration

Sep. 2009 – Aug. 2012 **Master of Science**, *ETH Zurich*, Zurich, Switzerland.
Major: Electrical Engineering and Information Technology
Thesis: "Compressed Sensing Reconstruction on FPGA"

Sep. 2005 – Jun. 2009 **Bachelor of Engineering**, *University of Electronic Science and Technology of China*, Chengdu, China.
Major: Integrated Circuit Design and Integrated System
Thesis: "Design of bus-based Hardware Accelerator System"

Professional Experience

May. 2020 – Aug. 2020 **Tech. Intern**, MATHWORKS, INC., Natick, United States.
- Object detection demo for Vision HDL Toolbox

May. 2019 – Aug. 2019 **Tech. Intern**, MATHWORKS, INC., Natick, United States.
- Image resizing demo for Vision HDL Toolbox
- Video Stabilization demo for Vision HDL Toolbox

May. 2018 – Aug. 2018 **Tech. Intern**, MATHWORKS, INC., Natick, United States.
- Feature detection and matching algorithm using Vision HDL Toolbox

Nov. 2012 – Aug. 2017 **Development Engineer**, NANOTRONIC GMBH, Lyss, Switzerland.
- FPGA based speed control system for high-speed trains
- FPGA based step motor controller for space application
- RaspberryPI based test system

Oct. 2014 – May. 2016 **FPGA Engineer**, ABB AG, Turgi, Switzerland.
- FPGA control hub for AD-DC converter
- FPGA based embedded system for data acquisition & processing

Feb. 2011 – Aug. 2011 **Test Engineer Intern**, JAQUET TECHNOLOGY GROUP, Basel, Switzerland.
- Design PCB for speed sensor IC
- Design UART control interface using Visual Basic
- Test and performance analysis

Professional Skills

Programming Language	VHDL, Verilog HDL, MATLAB, C, Python
Software	Xilinx ISE/Vivado, Actel Libro SoC, ModelSim/QuestaSim
FPGA	**Xilinx**: Spartan-series, Virtex-series, Zynq; **Actel**: RT ProASIC3; **Altera**: Cyclone II, Arria 10 SoC
Framework	TensorFlow, PyTorch

Selected Publications

- **L. Bai**, Y.C. Lyu, X.M. Huang,
"RoadNet-RT: High Throughput CNN Architecture and SoC Design for Real-Time Road Segmentationn",
arXiv preprint arXiv:2006.07644, 2020

- **L. Bai**, Y.C. Lyu, X.M. Huang,
"A Unified Hardware Architecture for Convolutions and Deconvolutions in CNN",
IEEE International Symposium on Circuits and Systems (ISCAS)
Seville, Spain, 11-14 October 2020

- **L. Bai**, Y.C. Lyu, X. Xu, X.M. Huang,
"PointNet on FPGA for Real-Time Lidar Point Cloud Processing",
IEEE International Symposium on Circuits and Systems (ISCAS)
Seville, Spain, 11-14 October 2020

- Y.C. Lyu, **L. Bai**, X.M. Huang,
"ChipNet: Real-Time LiDAR Processing for Drivable Region egmentation on an FPGA",
IEEE Transaction on Circuits and Systems I: Regular Papers, Vol.66 No.5, 2018

- **L. Bai**, Y.M. Zhao, X.M. Huang,
"A CNN Accelerator on FPGA using Depthwise Separable Convolution",
IEEE Transaction on Circuits and Systems II: Express Briefs, Vol.65 No.10, 2018

- H. Shen, C. Xu, Y. Yang, L. Sun, Z. Cai, **L. Bai**, E. Clancy, X. Huang
"Respiration and Heartbeat Rates Measurement Based on Autocorrelation Using IR-UWB Radar",
IEEE Transaction on Circuits and Systems II: Express Briefs, Vol.65 No.10, 2018

- Y.C. Lyu, **L. Bai**, X.M. Huang,
"Real-Time Road Segmentation Using LiDAR Data Processing on an FPGA",
IEEE International Symposium on Circuits and Systems (ISCAS),
Florence, Italy, 27-31 May 2018

- **L. Bai**, P. Maechler, M. Muehlberghuber, H. Kaeslin,
"High-Speed Compressed Sensing Reconstruction on FPGA Using OMP and AMP",
IEEE International Conference on Electronics, Circuits, and Systems (ICECS),
Seville, Spain, 9-12 Dec 2012

Interests

- Hiking
- Reading